

/ Requirement of Applying Gradient Descent Method in Multilevel Neuron Network :

A. activation function used in multilevel neuron network need to be differentiable.

B. sign function which usually use in a single perceptron is discontinuous at 0,
so sign function is indifferentiable and isn't fit for multilevel neuron network.

C. S-shape functions are continuous everywhere,
so S-shape functions are differentiable and are fit for multilevel neuron network.

there are two types of S-shape functions:

1. hyperbolic tangent function

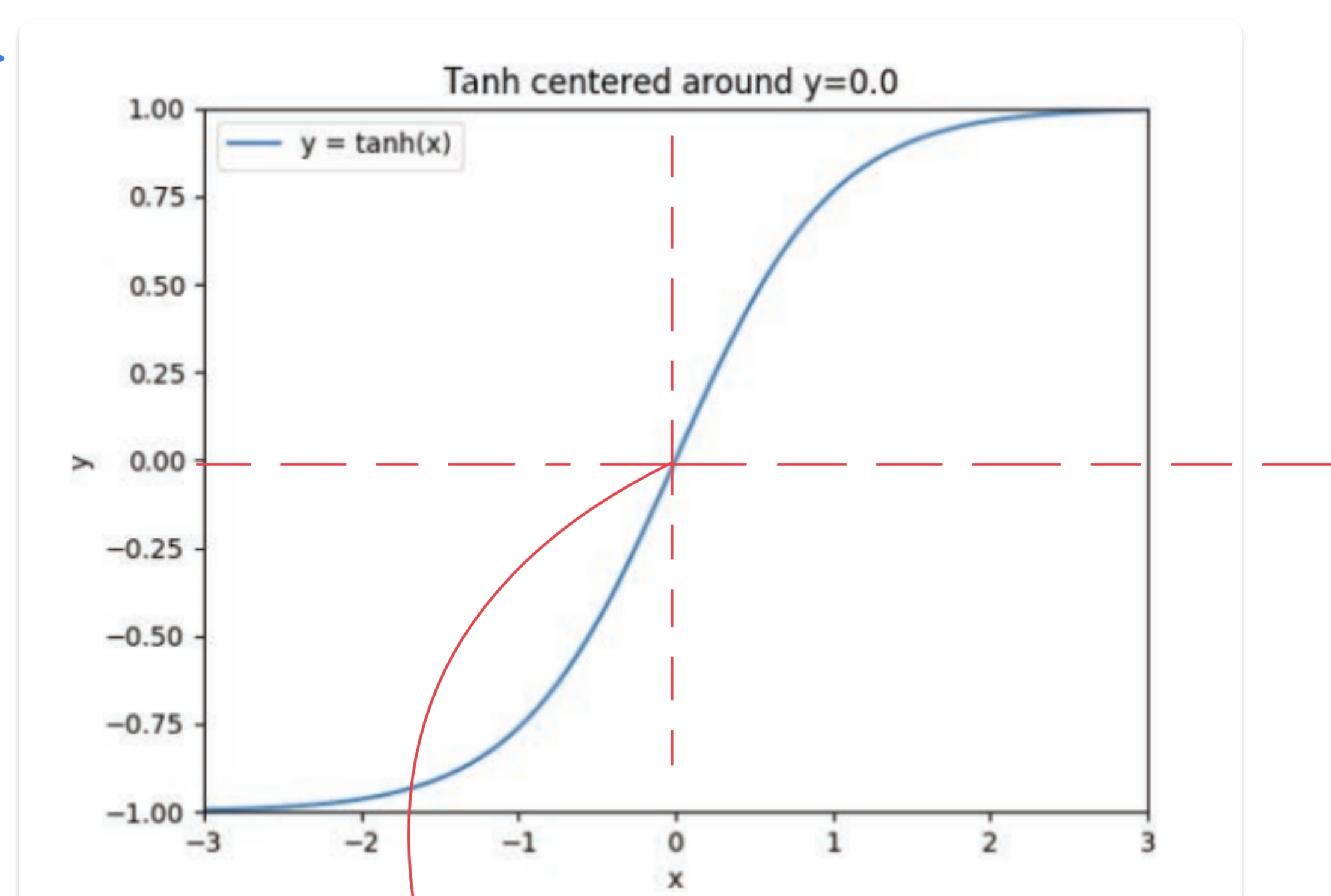
2. logistic function = sigmoid function = logistic sigmoid function

2. Introduction of Two types of S-shape Functions:

A. hyperbolic tangent function in "Mathematics":

$$\Rightarrow y = \tanh(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}} = \frac{e^{2x} - 1}{e^{2x} + 1}$$

$\Rightarrow$



threshold of $y = \tanh(x)$: $x=0$ & $y=0$

$$\Rightarrow y' = \tanh'(x) = 1 - (\tanh(x))^2$$

$\Rightarrow$ as the formula above, if the x value is the same,
derivative of hyperbolic tangent function ($\tanh'(x)$) can be computed only by hyperbolic tangent function $\tanh(x)$,
without using the x value directly.

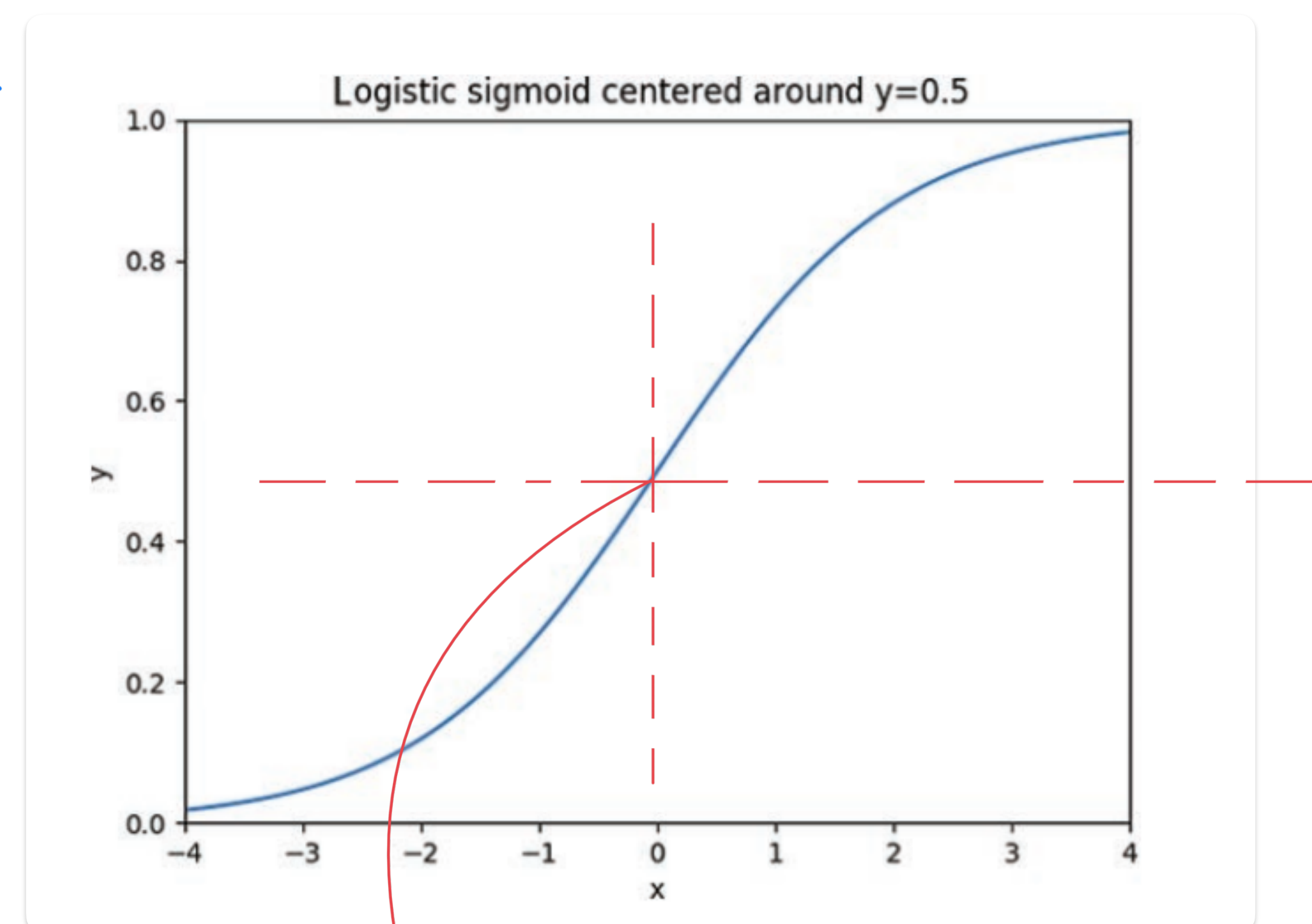
$\Rightarrow$ the variable x in "Mathematics" need to be replaced to function z (weighted sum function) in "Deep Learning"

the function y in "Mathematics" need to be replaced to function $\hat{y}$ (activation function of predicted output $\hat{y}$ from training perceptron) in "Deep Learning"

B. logistic sigmoid function in "Mathematics":

$$\Rightarrow y = S(x) = \frac{1}{1+e^{-x}} = \frac{e^x}{e^x + 1}$$

$\Rightarrow$



threshold of $y = S(x)$: $x=0$, $y=0.5$

$$\Rightarrow y' = S'(x) = S(x) \cdot (1 - S(x))$$

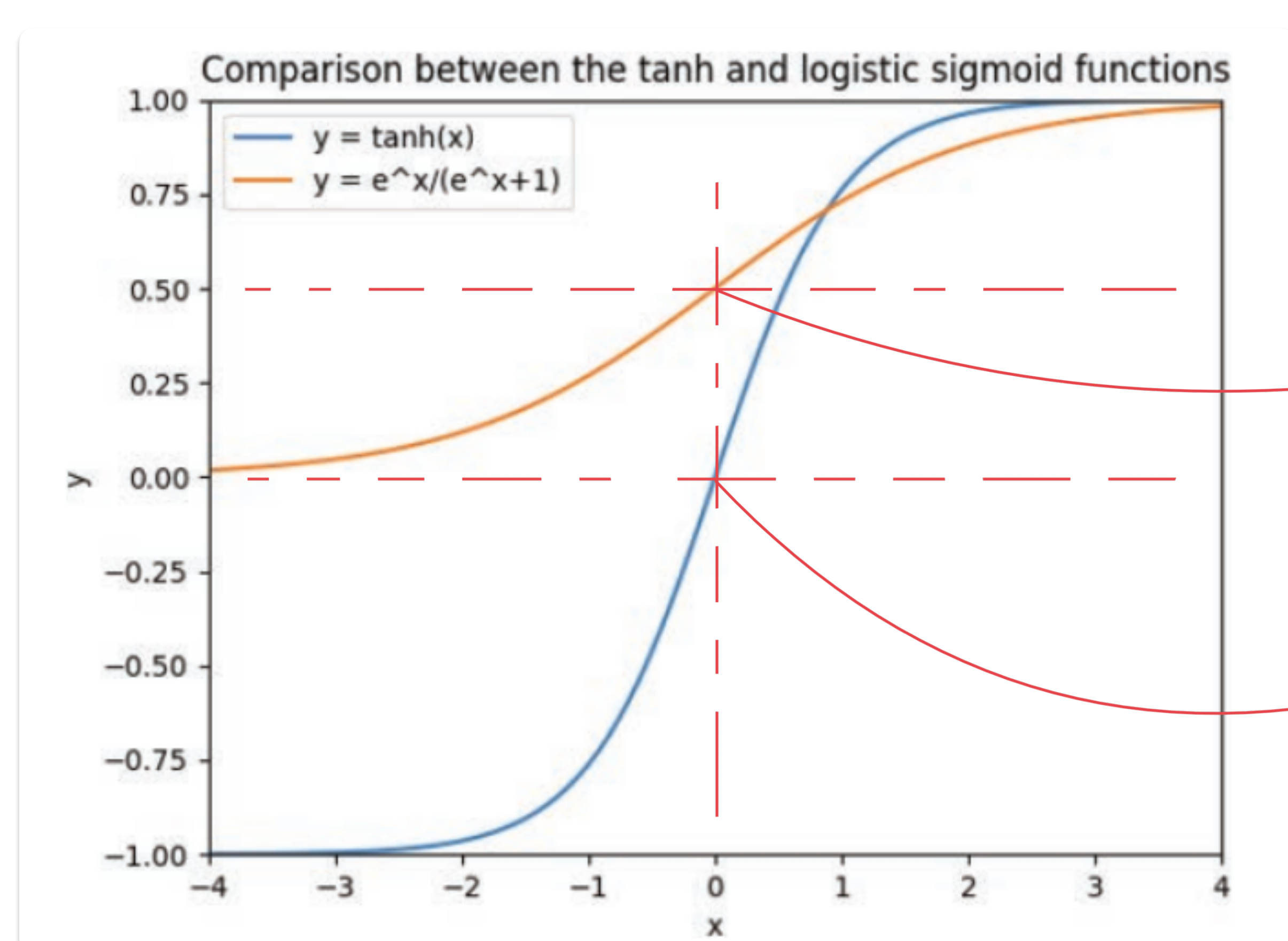
$\Rightarrow$ as the formula above, if the x value is the same,
derivative of logistic sigmoid function ($S'(x)$) can be computed only by logistic sigmoid function $S(x)$
without using the x value directly.

$\Rightarrow$ the variable x in "Mathematics" need to be replaced to function z (weighted sum function) in "Deep Learning"

the function y in "Mathematics" need to be replaced to function $\hat{y}$ (activation function of predicted output $\hat{y}$ from training perceptron) in "Deep Learning"

C. By adjusting bias weight (w_0) in weighted sum function z , we can use different w_0 value to decide what $w_1 \cdot x_1 + \dots + w_n \cdot x_n$ value can get z value = 0,
in this case, $w_1 \cdot x_1 + \dots + w_n \cdot x_n$ value corresponds to $\left| \begin{array}{l} \tanh(x) \text{'s threshold: } x=0, y=0 \\ S(x) \text{'s threshold: } x=0, y=0.5 \end{array} \right|$ can be changeable not fixed.

D.



threshold of $y = S(x)$: $x=0$, $y=0.5$

threshold of $y = \tanh(x)$: $x=0$ & $y=0$